12. If $\tan A - \tan B = x$ and $\cos B - \cos A = y$ prove that $\cot(A-B) = \frac{1}{x} + \frac{1}{y}$.

13. If $\sin\alpha + \sin\beta = a$ and $\cos\alpha + \cos\beta = b$, then prove that $\cos(\alpha-\beta) = \frac{a^2+b^2-2}{2}$.

14. Find the value of $\tan\frac{\pi}{8}$.

15. If $\tan\alpha = \frac{1}{7}$, $\sin\beta = \frac{1}{\sqrt{10}}$ Prove that $\alpha + 2\beta = \frac{\pi}{4}$ where $0 < \alpha < \frac{\pi}{2}$ and $0 < \beta < \frac{\pi}{2}$.

4.3 Transformation formulae

The two sets of transformation formulae are described below.

4.3.1 Transformation of the products into sum or difference

$$\sin A\cos B + \cos A\sin B = \sin(A+B) \quad (1)$$

$$\sin A\cos B - \cos A\sin B = \sin(A-B) \quad (2)$$

$$\cos A\cos B - \sin A\sin B = \cos(A+B) \quad (3)$$

$$\cos A\cos B + \sin A\sin B = \cos(A-B) \quad (4)$$

Adding (1) and (2), we get

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$

Subtracting (1) from (2), we get

$$2\cos A\sin B = \sin(A+B) - \sin(A-B)$$

Adding (3) and (4), we get

$$2\cos A\cos B = \cos(A+B) + \cos(A-B)$$

Subtracting (3) from (4), we get

$$2\sin A\sin B = \cos(A-B) - \cos(A+B)$$

Thus, we have the following formulae;

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$

$$2\cos A\sin B = \sin(A+B) - \sin(A-B)$$

$$2\cos A\cos B = \cos(A+B) + \cos(A-B)$$

$$2\sin A\sin B = \cos(A-B) - \cos(A+B)$$

Also

$$\sin^2 A - \sin^2 B = \sin(A+B)\sin(A-B)$$

$$\cos^2 A - \sin^2 B = \cos(A+B)\cos(A-B)$$

4.3.2 Transformation of sum or difference into product

$$\sin C + \sin D = 2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$$

$$\sin C - \sin D = 2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$$

$$\cos C + \cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$$

$$\cos C - \cos D = -2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$$

Example 4.14

Express the following as sum or difference

(i) $2\sin 2\theta\cos\theta$ (ii) $2\cos 3\theta\cos\theta$

(iii) $2\sin 4\theta\sin 2\theta$ (iv) $\cos 7\theta\ \cos 5\theta$

(v) $\cos\frac{3A}{2}\cos\frac{5A}{2}$ (vi) $\cos 9\theta\sin 6\theta$

(vii) $2\cos 13A\sin 15A$

Solution

(i) $2\sin 2\theta\cos\theta = \sin(2\theta+\theta) + \sin(2\theta-\theta)$
$= \sin 3\theta + \sin\theta$

(ii) $2\cos 3\theta\cos\theta = \cos(3\theta+\theta) + \cos(3\theta-\theta)$
$= \cos 4\theta + \cos 2\theta$

(iii) $2\sin 4\theta\sin 2\theta = \cos(4\theta-2\theta) - \cos(4\theta+2\theta)$
$= \cos 2\theta - \cos 6\theta$

(iv) $\cos 7\theta\ \cos 5\theta = \frac{1}{2}[\cos(7\theta+5\theta) + \cos(7\theta-5\theta)]$
$= \frac{1}{2}[\cos 12\theta + \cos 2\theta]$

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